ELE 709:
Real-time Computer Control Systems

Lecture 9

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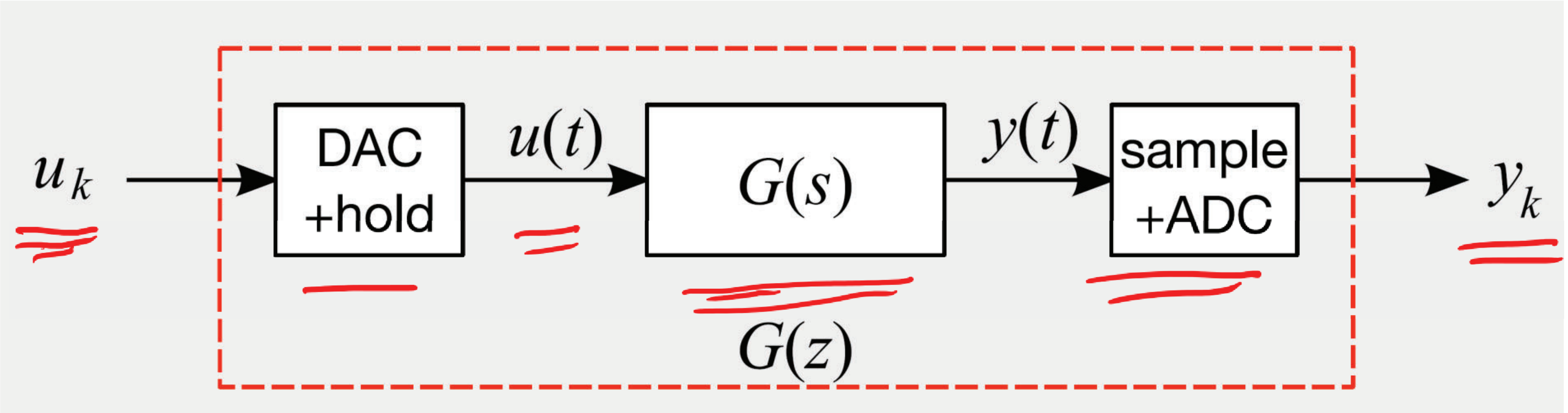
Contents:

- Other aspects in digital control:
  - Zero-order hold
  - Quantization noise
  - Sampling effect

Digital control



- A/D or ADC: Analog-to-Digital Convertor
- D/A or DAC: Digital-to-Analog Convertor

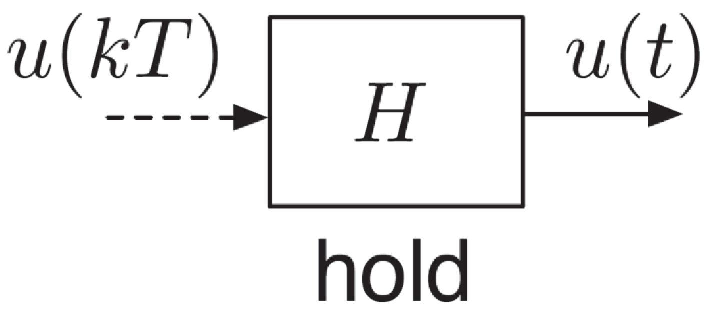
Zero-order Hold (ZOH)

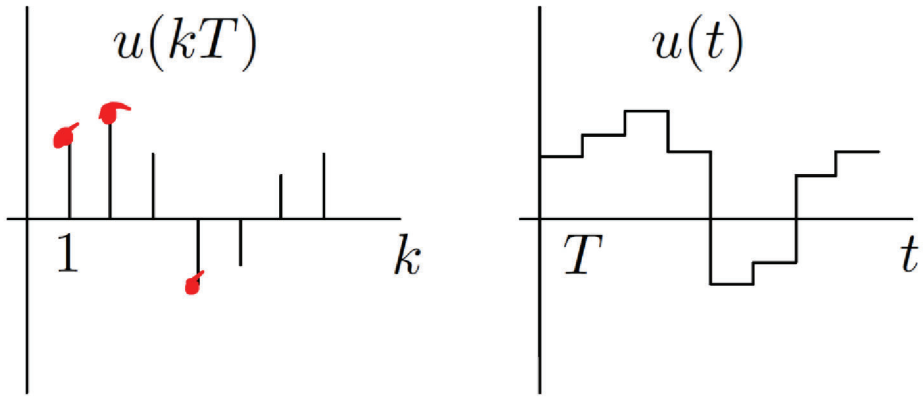
- Definition of the output of ZOH:

$$\boxed{u(t) = u(kT),} \qquad kT \leq t < (k+1)T$$

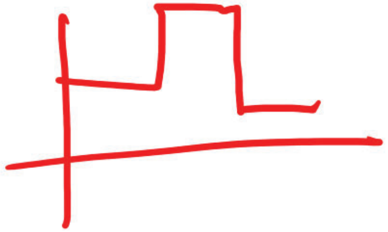
- converts the discrete input into a continuous-time signal

- part of the D/A



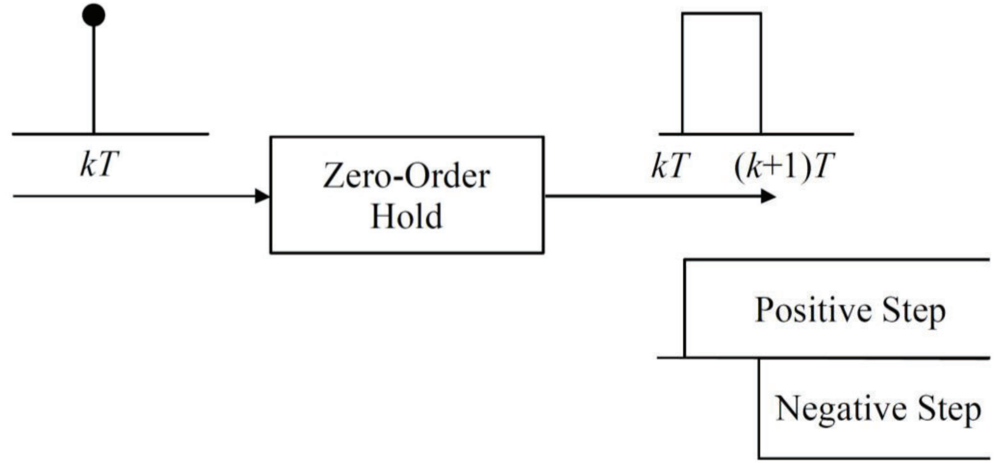


Zero-order Hold (ZOH)

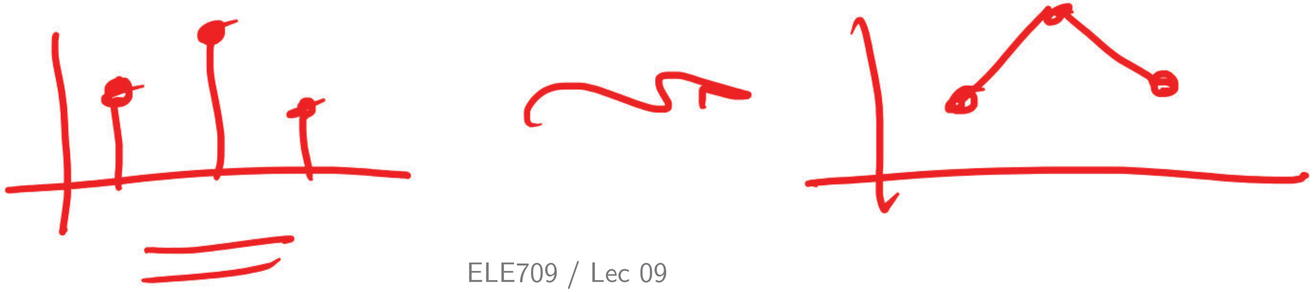


- Transfer function of ZOH:

$$G_{\mathsf{ZOH}}(s) = \frac{1 - e^{-sT}}{s}$$

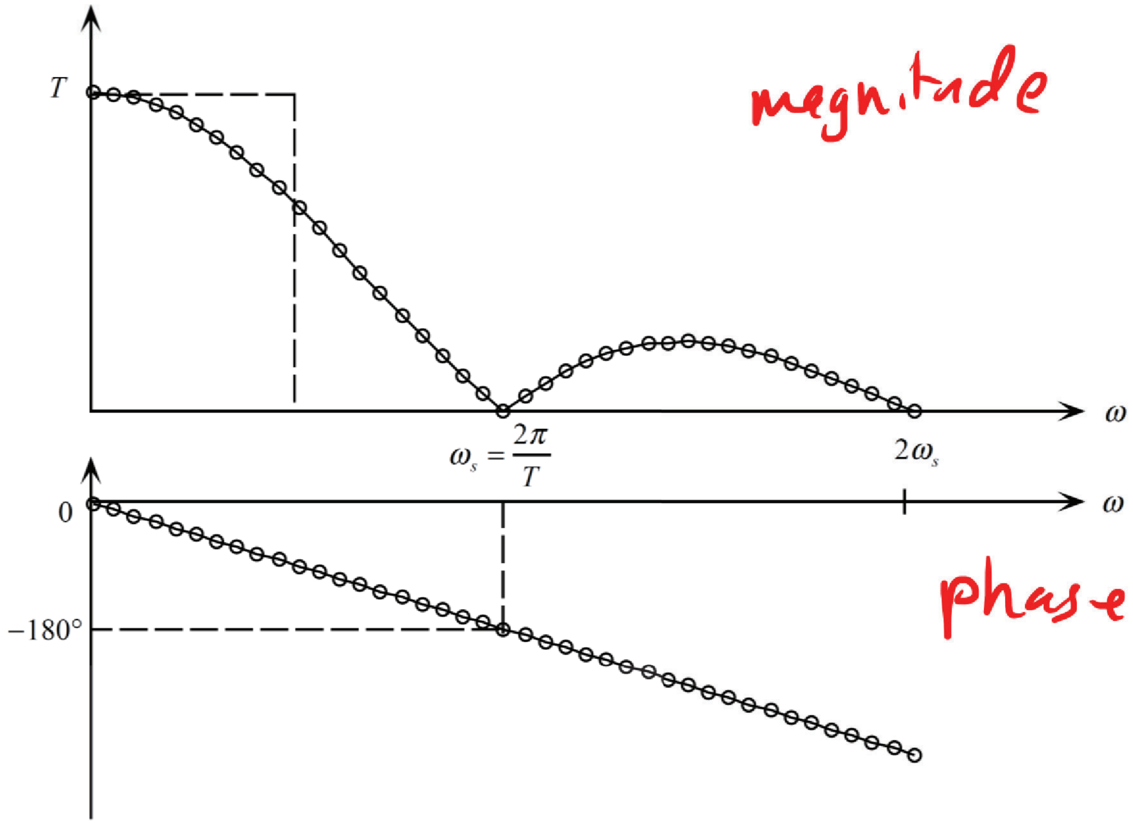


1st-order hold

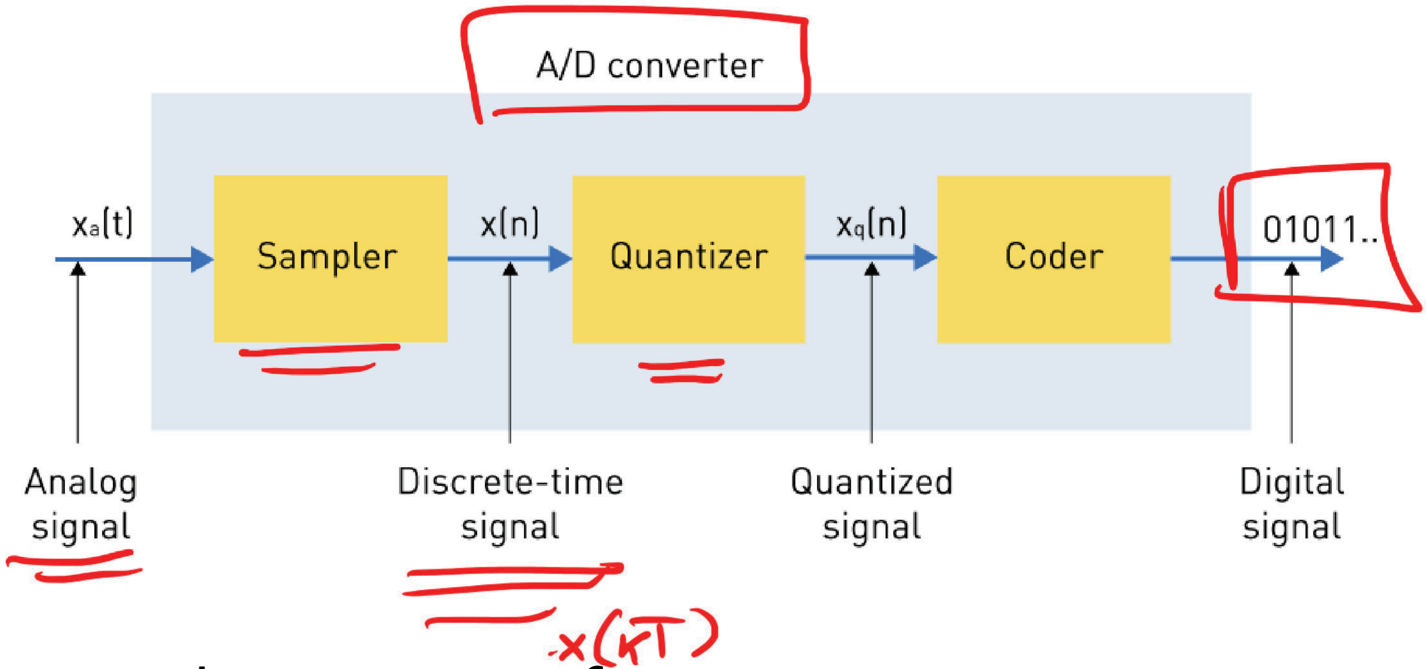


Zero-order Hold (ZOH)

- Frequency response of ZOH:
    - The ZOH is a low-pass filter,
    - and has linear phase lag with frequency.
- This introduces a delay in the control loop which may have a destabilizing effect for large sampling periods.



Quantization effects



- Quantization is the process of converting a continuous range of sampled signal values to a finite set of discrete levels (digital word).

Quantization effects

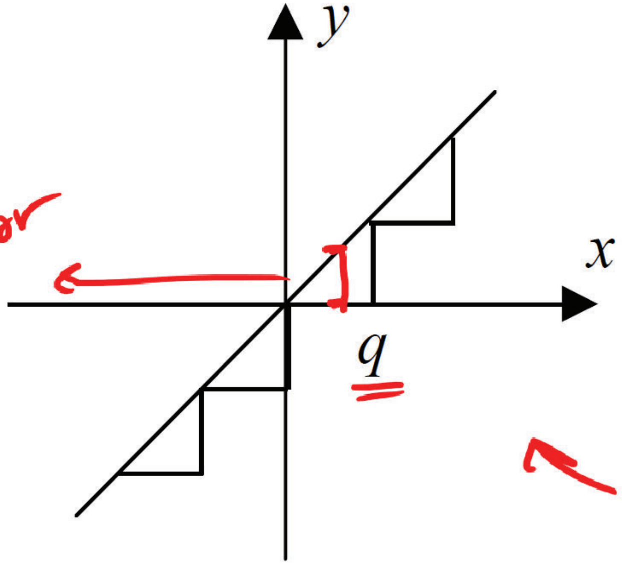
- The smaller the A/D resolution, the better the performance.
- The cost of a component increases as the word length increases.
- Depending on the choice of the A/D, the word length is mainly determined by the quantization effects.

- Example,
  - A sensor has a 5 mV noise and a 5 V range.
  - No need for a A/D with more than 10 bits.
  - Resolution of having 2^{10} levels corresponds to an error of 0.1%, which is equal to the noise level.

Quantization noise/error

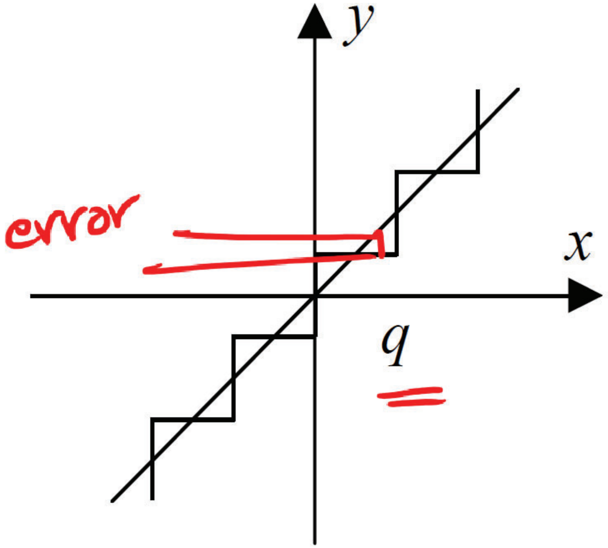
- Effects of the quantization due to A/D rounding or truncation.

error



truncation

error



rounding

n: number of bits

q is the quantization level, i.e. the range of A/D divided by 2^n

→ e.g. [−1, 1]   −1, −0.5, 0, .5, 1

Quantization noise/error

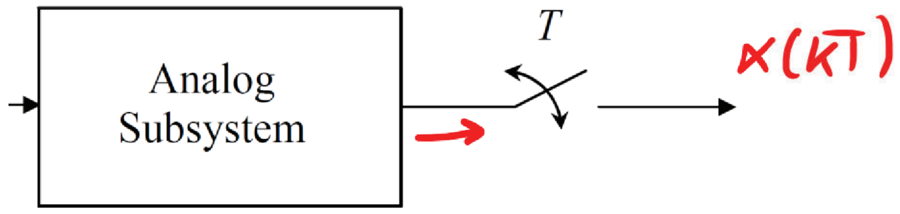
- The noise/error due to quantization can be modeled as a uniformly distributed random process.

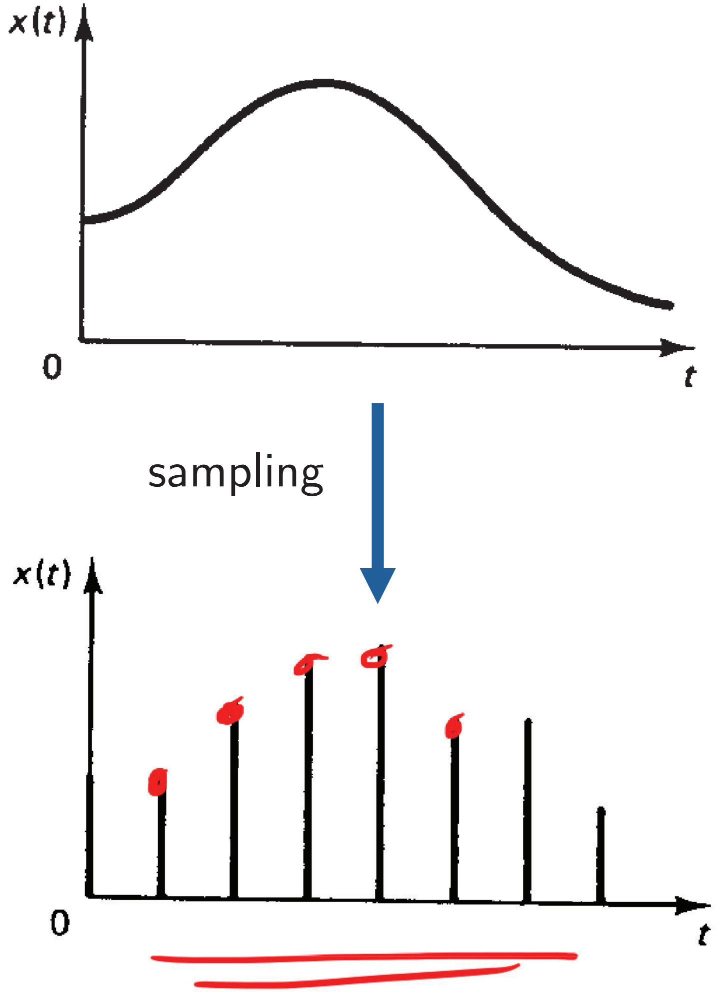
  - Rounding: mean $= 0$ variance $= \dfrac{q^2}{12}$
  - Truncation: mean $= \dfrac{q}{2}$ variance $= \dfrac{q^2}{12}$

- Effect of the quantization noise/error can be reduced by decreasing the sampling period, although it requires more expensive hardware and may aggravate problems caused by the finite-word representation of parameters.

Sampling considerations

- We consider we have a A/D with an ideal sampler with sampling period T.
- In practice, quantization errors are present, although small but nonzero.
- Also, variations in the actual sampling rate exist but are negligible.





Sampling considerations


Claude Shannon


Harry Nyquist

- Sampling Theorem (Nyquist–Shannon Theorem):

We can recover a signal from its samples if

$$\omega_s > 2\omega_b$$

where $\boxed{\omega_s = \frac{2\pi}{T}}$

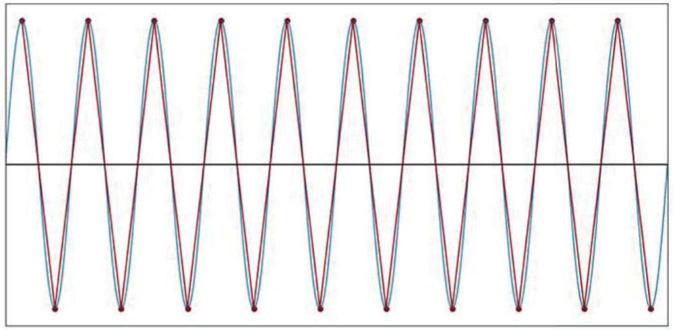
ω_s : sampling frequency (rad/s)

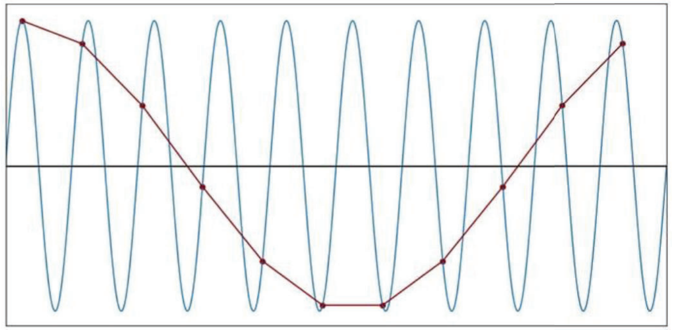
ω_b = highest frequency in the original signal (bandwidth)

(rad/s)

$F = \frac{1}{T}$ (Hz)

1 Hz $= 2\pi$ rad/s


2 samples per period


1.1 samples per period

Sampling considerations

- For first-order systems $G(s) = \dfrac{K}{\tau s + 1}$, then $\omega_b = \tau^{-1}$

- For second-order systems $\boxed{G(s) = \dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}}$

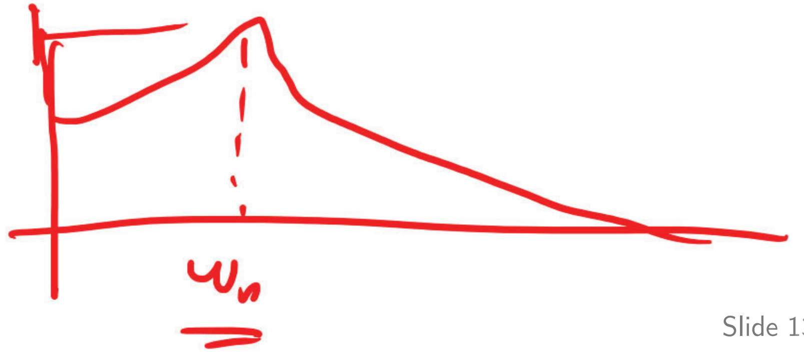
ω_n : natural freq.

ζ : damping factor

then $$\omega_b = \omega_n\sqrt{1 - 2\zeta^2 + \sqrt{2 - 4\zeta^2 + 4\zeta^4}}$$

- For higher-order systems

$$\boxed{\omega_b \approx \omega_n}$$



Sampling considerations

- In practice, the sample frequency is chosen to be much larger than (at least twice) the bandwidth of the system.

- As a rule of thumb, for a smooth operation we need:

$$\boxed{20 \leq \frac{\omega_s}{\omega_b} \leq 40}$$

- In practice, the sampling frequency should be upper-bounded.

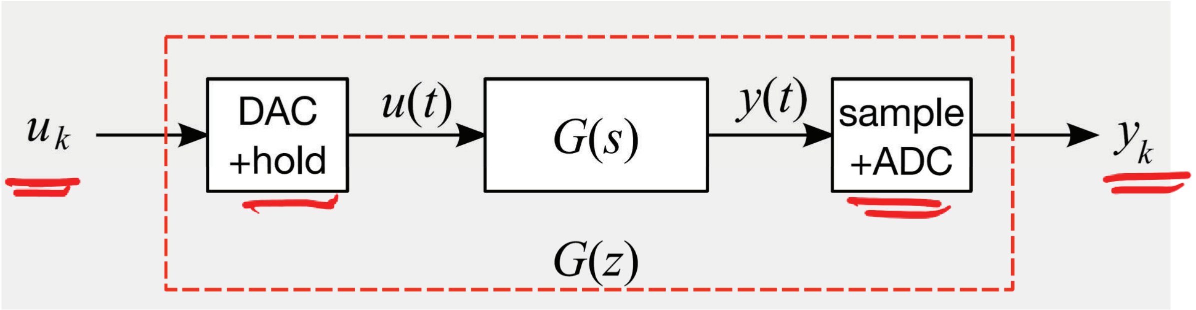
Latency

- Suppose that the D/A and A/D are not synchronized, and that there is a fixed delay $\boxed{T_C}$ between the sampling of the system output measurement at the A/D and the updating of the D/A.

$$0 \leq T_C \leq T$$

- Possible reasons:
  - Computation/processing delay
  - Actuator/sensor delay

Digital control



- Combination of A/D, D/A and the continuous-time system:

$$G_{\text{combined}}(z) = (1 - z^{-1})\mathcal{Z}\left\{\mathcal{L}^{-1}\left\{\frac{G(s)}{s}\right\}\right\}$$

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